PORTFOLIO PROJECT · HEALTHCARE FINANCE · PYTHON

# Cancer service line profitability

Which parts of a hospital cancer program make money after all costs, what that result depends on, and where AI helps. All deliverables on one page.

> Real hospital cost data is proprietary, so every figure comes from a synthetic dataset (3,600 patients, 79,977 encounters, fiscal 2025, seed 42) built from stated assumptions. The patterns show how the analysis works. The dollar amounts are not benchmarks.

| $17.0M | +$27.7M | −$0.6M | 47.3% |
|---|---|---|---|
| Operating margin, 8.1% of $210.7M net revenue | Commercial payers. All other payers combined: −$10.6M | Margin if the 340B drug discount ($17.7M) disappears | Of patients have a negative operating margin |

## Deliverables

| DELIVERABLE | WHAT IT IS | WHERE |
|---|---|---|
| Dashboard | Interactive view of the results, with all 11 charts in a gallery | `claude.ai/artifact/LruywhrjTxNifaQ3Gt5jjd` |
| Case study | Problem, method, findings and AI, with ten charts beside the text | `claude.ai/code/artifact/5abf673e-156d-4b53-bba1-024e91d8e3ef` |
| Slide deck | 13-slide story with five chart slides | `claude.ai/artifact/FNGN3wXUxg37BX2MzgavrT` |
| Project zip | Code, synthetic data, outputs, charts and README | `cancer-service-line-profitability.zip` |
| Data generator | Synthetic patients and encounters; every assumption is a constant at the top | `src/generate_data.py` |
| Analysis | P&L, cuts by payer, cancer type and service, drug economics, denials, scenarios | `src/analyze.py, outputs/` |
| Integrity test | Recomputes the P&L from raw rows and ties every table to it | `tests/check_integrity.py` |
| Denial-risk model | Gradient boosting model and review-worklist value | `src/denial_model.py` |
| Charts | 11 PNG charts, one finding each | `src/make_charts.py, charts/` |
| README | Problem statement, method, run instructions, charts, takeaways | `README.md` |

## The problem

Cancer programs are among the most expensive and least transparent service lines in a hospital. Leaders must decide where to invest and which payer contracts to fight for, but margin is hard to see. Drug margin hides inside pass-through pricing, overhead is spread by rule of thumb, and money leaks through denials and underpayments.

In this project's data the program earns $17.0M. That rests on commercial payers and one drug discount, nearly half of patients lose money, and cancer type rankings change with the

overhead method. A leader who sees only the total cannot tell which contracts, services or risks matter.

| FINDING | VALUE |
|---|---|
| Net patient revenue | $210.7M |
| Operating margin | $17.0M (8.1%) |
| Commercial payers | +$27.7M |
| All other payers combined | −$10.6M |
| Operating margin without the 340B drug discount | −$0.6M |
| Patients with a negative operating margin | 47.3% |
| Cancer type and payer combinations that lose money | 28 of 40 |
| Inpatient admissions / CAR-T cell therapy | −$4.7M / −$1.4M |
| Prostate margin: overhead by non-drug cost vs by net revenue | −2.1% vs 6.4% |

- **Commercial payers carry the line.** Medicare Advantage, Medicaid and self-pay all lose money in total, and Medicare fee-for-service is roughly break-even.
- **Payer mix is the strongest pattern across cancer types.** Prostate (73% Medicare or Medicare Advantage) loses 2.1%. Breast (53% commercial) earns 14.5%.
- **Drugs decide the result.** Infusion drug margin is $19.7M with the modeled 340B discount and $2.1M without it.

## The solution

Build the P&L from encounter-level data, tie every table back to the raw rows, then cut it by payer, cancer type, service and drug economics, and stress test it.

| STEP | HOW EACH ENCOUNTER IS TREATED | TOTAL |
|---|---|---|
| Expected payment | Gross charge less contractual adjustment | $231.1M |
| Net revenue | Less underpayments ($1.6M), denial write-offs after appeals ($10.7M), bad debt and charity ($8.2M) | $210.7M |
| Direct margin | Less drug acquisition, pharmacy handling, clinical labor, supplies, equipment and facility, denial rework ($172.6M) | $38.1M |
| Operating margin | Less allocated overhead ($21.0M), spread on non-drug direct cost because drugs are a pass-through | $17.0M |

### Five moves the analysis points to

| | |
|---|---|
| −$17.7M | **Plan for losing 340B.** The largest single risk, and the most assumption-sensitive result in the project. |
| +$5.3M | **Renegotiate the weakest commercial contracts.** A 5% commercial rate increase. |
| +$2.7M | **Recover denied claims.** Recovering 30% of write-offs, after a 15% cost to collect. |
| −$6.1M | **Fix inpatient and CAR-T economics.** Current losses of $4.7M and $1.4M. |
| −$5.2M | **Prepare for site-neutral payment.** The effect if all payers follow. |

Do not cut a service line on one number. Compare overhead methods first, because the ranking of cancer types changes with the method.

## How AI helps

AI sharpens decisions and recovers leakage. It does not change payer mix or 340B policy.

| PROBLEM IN THE DATA | AI APPROACH | SIZE HERE |
|---|---|---|
| Rankings change with the overhead method | Time-driven costing: models chair, linear accelerator and operating room minutes from electronic health record timestamps instead of a flat percentage | −2.1% vs 6.4% |
| Claim denials | A model scores each claim for denial risk before submission; a language model drafts appeals from the chart | $10.7M written off |
| Underpayments | An engine compares each remittance to contract terms line by line | $1.6M |
| Dependence on 340B and payer mix | Forecasting and scenario models that update with volume, payer mix and policy | −$26.3M combined |
| Which patients lose money | Predict expected cost and coverage at treatment planning, to guide financial counseling and contract targets | 47% negative |

## The one approach built here: a denial-risk model

A gradient boosting classifier uses only information known before submission (payer, service, cancer type, stage, care setting, expected payment, month) and ranks claims for review. The split is by patient, so the 23,925 test claims (8.4% denied) come from patients the model never saw.

| REVIEW LIST: TOP 20% OF CLAIMS | DENIALS CAUGHT | NET VALUE PER YEAR |
|---|---|---|
| Random | 19% | $0.4M |
| Payer-and-service lookup | 43% | $1.7M |
| Gradient boosting model | 43% | $1.7M |
| Gradient boosting, ranked by expected dollars | 28% | $3.0M |

- Ranking by risk finds denials at 2.2 times the rate of random review (18.1% precision against 8.4%).
- **The model does not beat the simple payer-and-service lookup** (ROC AUC 0.704 against 0.706). The synthetic denials come from payer and service rates, so there is nothing more to learn. Real gains would come from claim-level features such as authorization status, codes and documentation flags. Start with the simple rule, and add the model when those features exist.

- Ranking by expected dollars catches fewer denials but recovers more money, which is how a real worklist should be ordered.
- Net value assumes 40% of reviewed would-be denials are fixed in time, less $25 per claim reviewed. Both are constants at the top of the script.
- Patient-level predictions belong in financial counseling and contracting, never in steering patients away from care. Payers use AI to deny claims too, so gains can turn into an arms race. The other approaches in the table are described, not implemented.

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# Charts

Each chart states its finding in the title. Grey is context, blue is money earned, and orange is money lost or at risk.

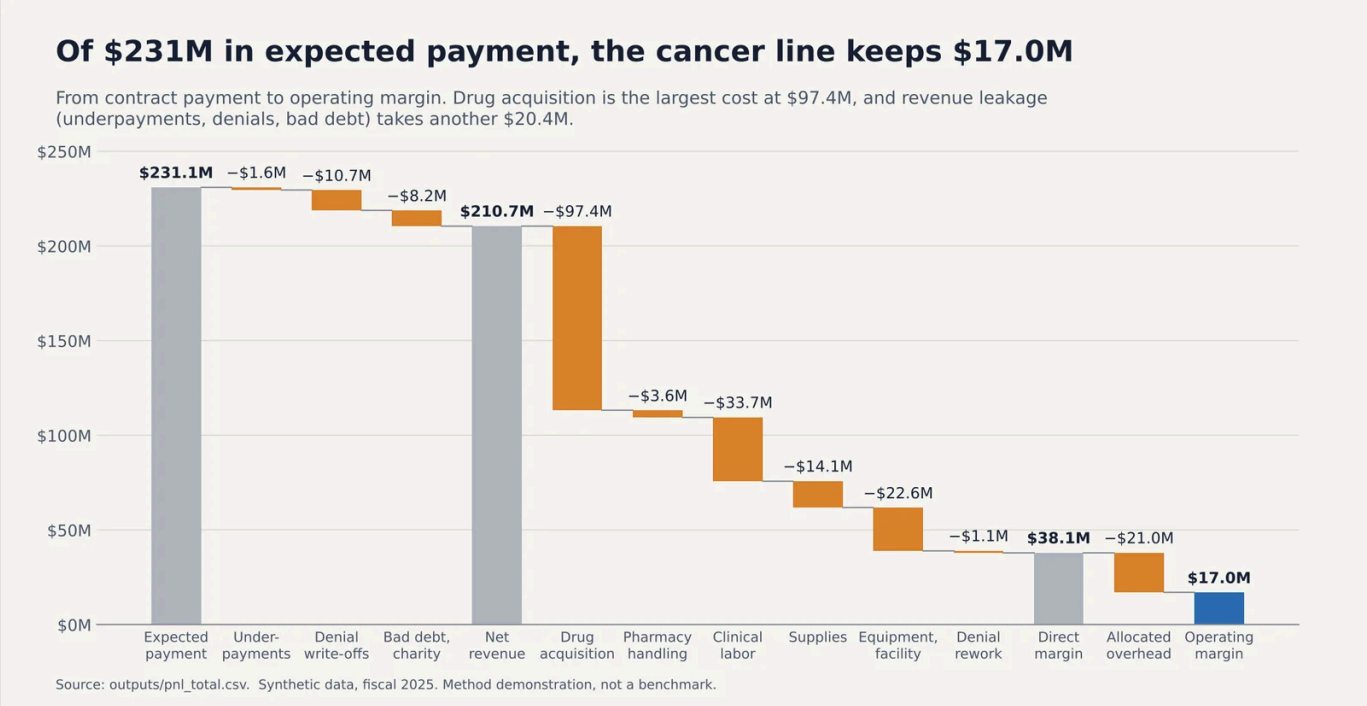
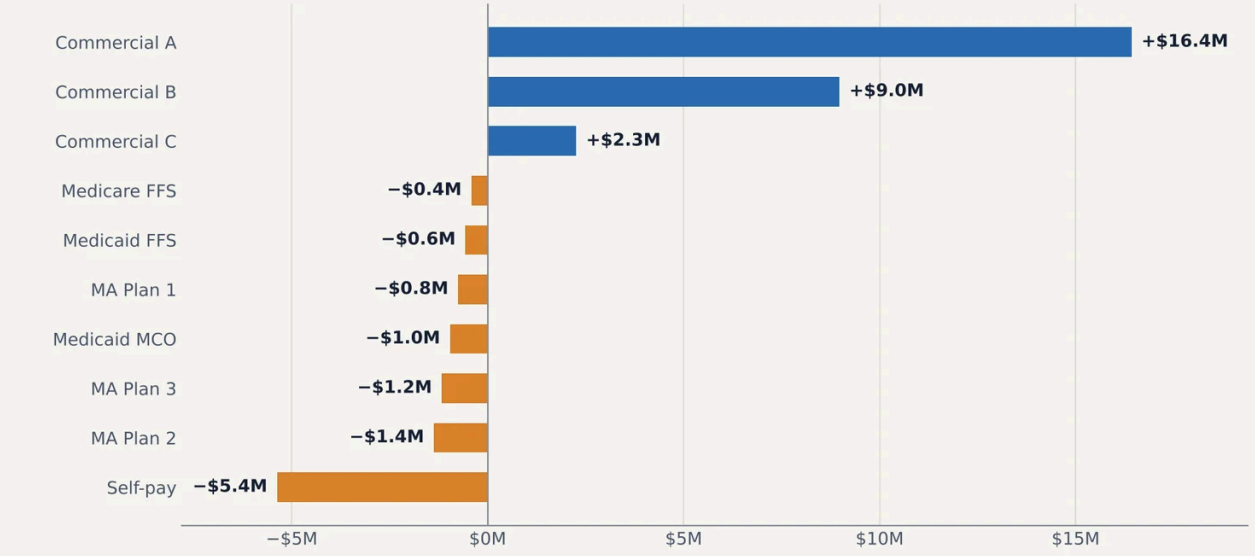



Chart 1 · Revenue waterfall · source outputs/pnl_total.csv · synthetic data

## Only the three commercial plans make money

Operating margin by payer, fiscal 2025. Commercial earns +$27.7M; every other payer group together loses $10.6M. Self-pay is the largest single loss.

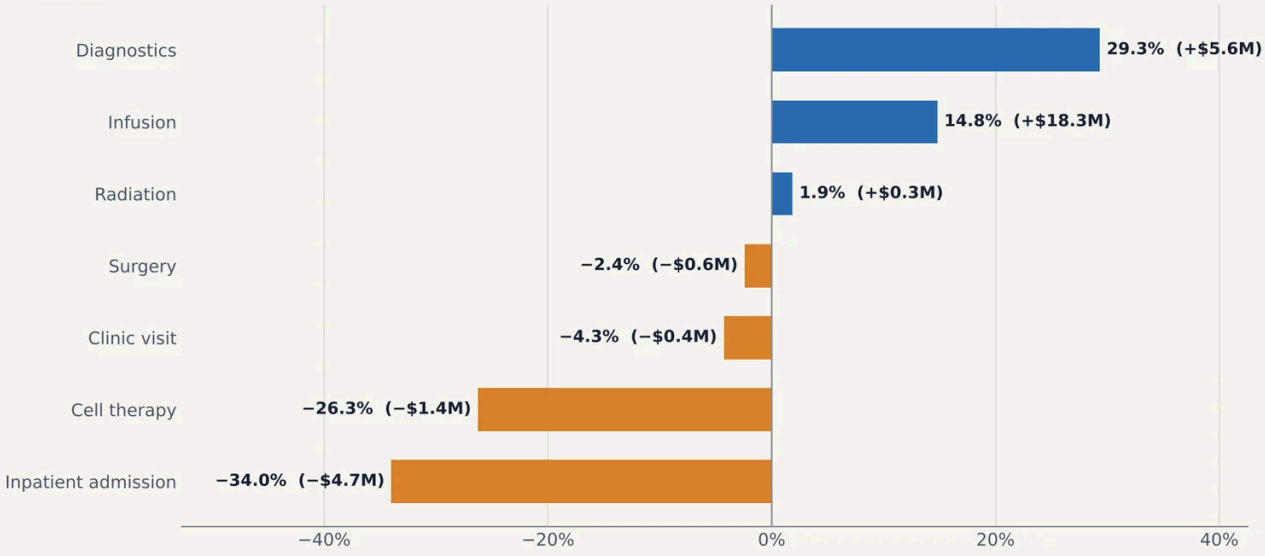



Source: outputs/by_payer_name.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 2 · Margin by payer · source outputs/by_payer_name.csv · synthetic data

## Diagnostics and infusion carry the line; inpatient and CAR-T lose

Operating margin as a percent of net revenue by service, with dollars in brackets. Infusion earns most of its margin from drugs, which depends on the 340B discount.




Source: outputs/by_service.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 3 · Margin by service · source outputs/by_service.csv · synthetic data

## Only commercial pays its way for every cancer type

28 of 40 cancer-by-payer groups lose money. Operating margin as a percent of net revenue; colors cap at ±40%, and self-pay losses are far beyond that scale.

| | Commercial | Medicare FFS | Medicare Advantage | Medicaid | Self-pay |
|---|---|---|---|---|---|
| Breast | 29% | −2% | −9% | −22% | < −100% |
| Gynecologic | 25% | −6% | −23% | −18% | < −100% |
| Lung | 28% | 6% | −2% | 1% | < −100% |
| Other | 25% | 4% | −4% | −13% | < −100% |
| Hematologic | 22% | −3% | −12% | 7% | < −100% |
| Colorectal | 27% | −6% | −20% | −41% | < −100% |
| Pancreatic | 22% | −11% | −23% | −73% | < −100% |
| Prostate | 30% | −10% | −15% | −61% | < −100% |

Source: outputs/cancer_by_payer.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 4 · Cancer type by payer · source outputs/cancer_by_payer.csv · synthetic data

## Cancer type rankings change with the overhead method

Operating margin % under four ways of allocating overhead. Prostate runs from −2.1% to 6.4% and pancreatic from 0.4% to 7.1%, while breast stays high under every method.

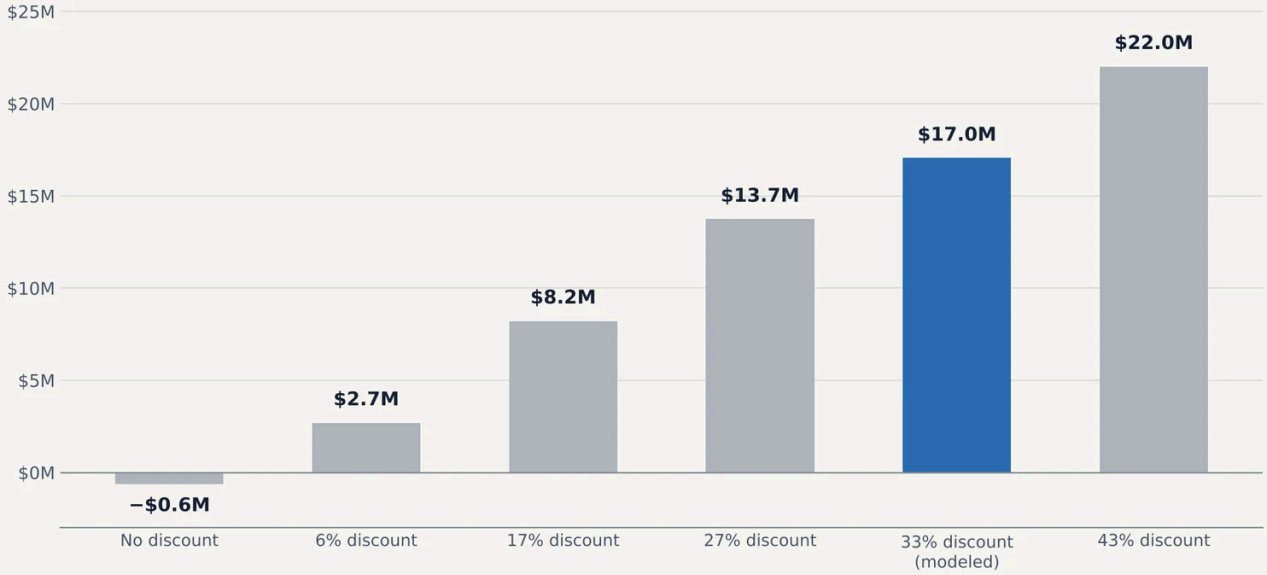



Source: outputs/overhead_sensitivity.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 5 · Overhead sensitivity · source outputs/overhead_sensitivity.csv · synthetic data

## Without the 340B drug discount the margin disappears

Operating margin by size of the discount on infusion drugs. Losing the modeled discount costs $17.7M, more than the whole margin. The discount size is an assumption.

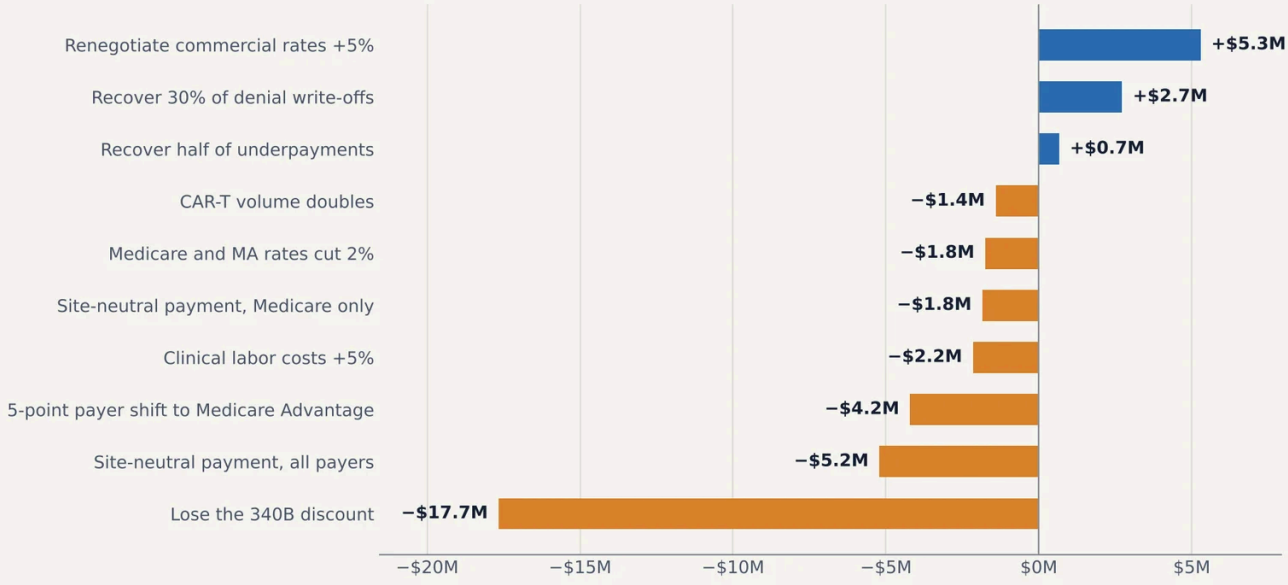



Source: outputs/sensitivity_340b.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 6 · 340B sensitivity · source outputs/sensitivity_340b.csv · synthetic data

## Losing the 340B discount is the largest risk

Change in annual operating margin by scenario. The combined downside (payer shift, 340B, Medicare site-neutral, labor +5%) is −$26.3M, taking margin from 8.1% to −4.5%.

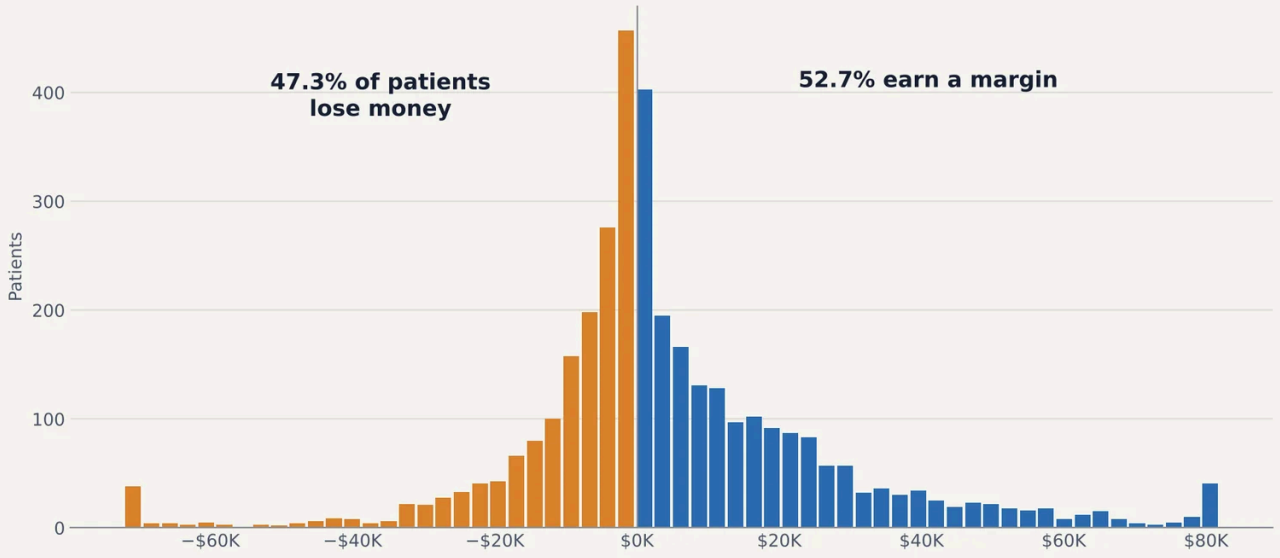



Source: outputs/scenarios.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 7 · Stress scenarios · source outputs/scenarios.csv · synthetic data

## Nearly half of cancer patients cost more than they bring in

Operating margin per patient (3,600 patients). Median +$0.4K, mean +$4.7K: a few high-margin patients carry the rest. Axis trimmed to the 1st to 99th percentile.

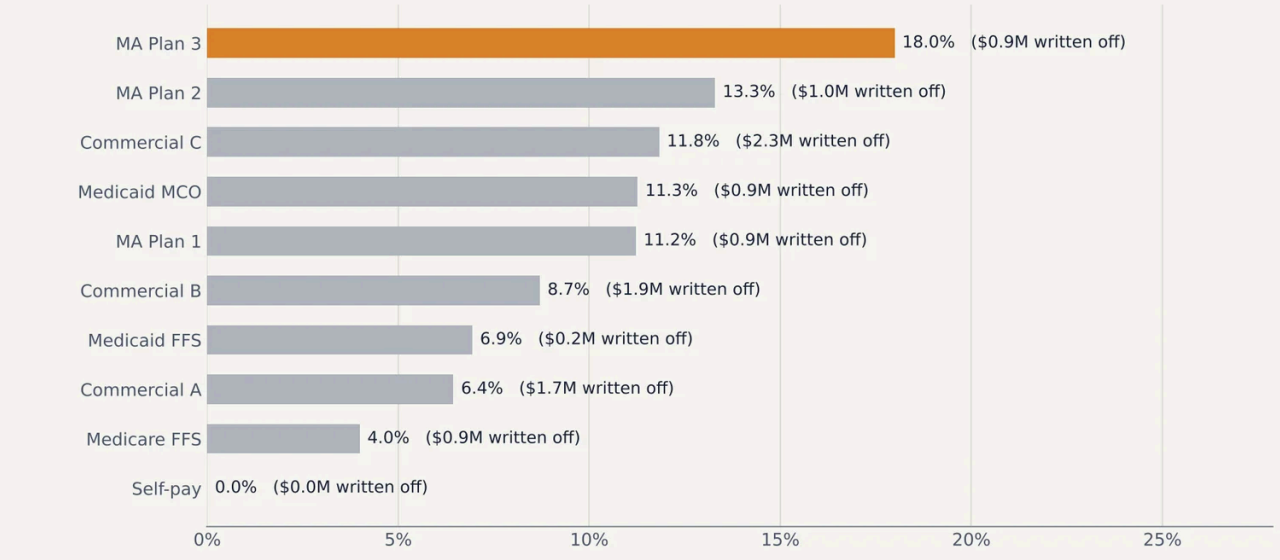



Source: outputs/patient_level_margin.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 8 · Patient margin distribution · source outputs/patient_level_margin.csv · synthetic data

## MA Plan 3 denies the most claims at first pass

Share of claims denied, with the dollars written off after appeals. Total denial write-offs are $10.7M.

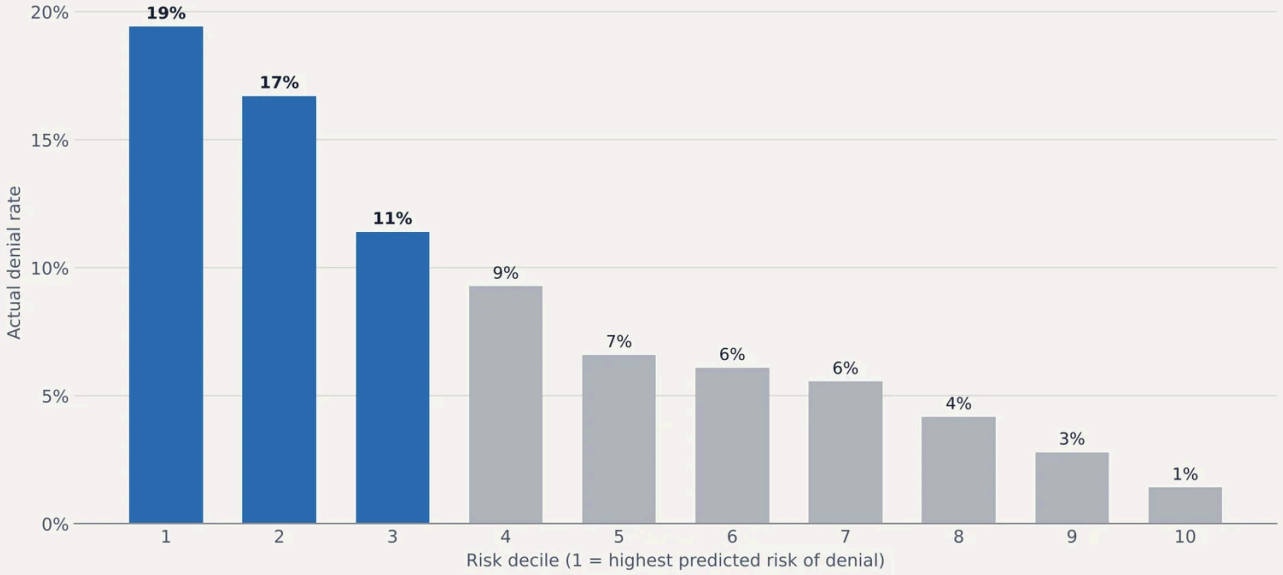



Source: outputs/denials_by_payer.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 9 · Denials by payer · source outputs/denials_by_payer.csv · synthetic data

## The denial model sorts claims well but matches a simple rule

Actual denial rate by predicted-risk decile on held-out claims. The top three deciles hold 68% of avoidable cost. The model matches the simple baseline, so use the baseline first.

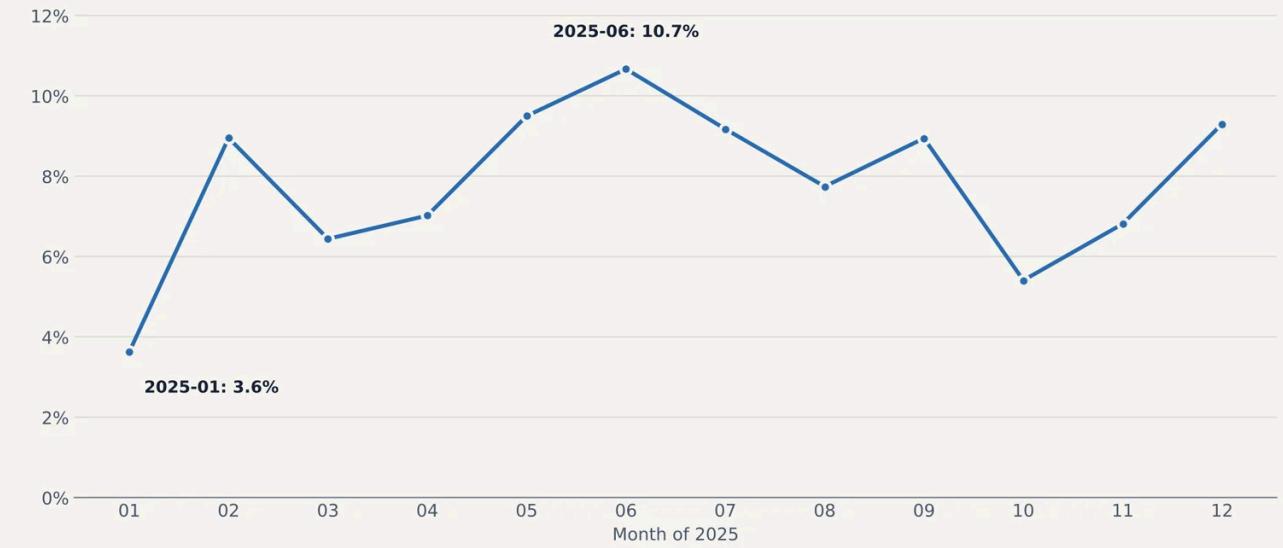



Source: outputs/denial_model_deciles.csv, outputs/denial_model.json. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 10 · Denial model deciles · source outputs/denial_model_deciles.csv · synthetic data

## Monthly margin swings widely around the 8.1% full-year result

Operating margin as a percent of net revenue by month. Month-to-month volatility is why a single month is a poor basis for a service line decision.




Source: outputs/monthly.csv. Synthetic data, fiscal 2025. Method demonstration, not a benchmark.

Chart 11 · Monthly margin · source outputs/monthly.csv · synthetic data

## Scenarios

Change in operating margin from the $17.0M base, from `outputs/scenarios.csv`.

| SCENARIO | CHANGE |
|---|---|
| Lose the 340B discount | −$17.7M |
| Combined downside (payer shift, no 340B, Medicare site-neutral, labor +5%) | −$26.3M |
| Shift 5 points of patients from commercial to Medicare Advantage | −$4.2M |
| Site-neutral payment, all payers | −$5.2M |
| Site-neutral payment, Medicare only | −$1.8M |
| Medicare and Medicare Advantage rates cut 2% | −$1.8M |
| Clinical labor +5% | −$2.2M |
| CAR-T volume doubles | −$1.4M |
| Renegotiate commercial rates +5% | +$5.3M |
| Recover 30% of denial write-offs (15% cost to collect) | +$2.7M |
| Recover 50% of underpayments (15% cost to collect) | +$0.7M |

## Assumptions and limitations

### Key assumptions

- Medicare pays drugs at 104.3% of a benchmark price. Commercial plans pay 112% to 142%.
- Drugs for 340B-eligible patients cost 64% of the benchmark, against 96% for others. **This is the most assumption-sensitive result.**
- Commercial rates run about 1.3 to 2.5 times Medicare. Medicaid pays 68% to 75% of Medicare.
- First-pass denial rates run from 4% to 18%, with appeal win rates of 57% to 78%.
- CAR-T payment is a multiple of hospital cost, below 1.0 for government payers and the weakest commercial contract.

### Limitations

- Single year, with no payment timing, appeals lag or seasonality.
- Encounter costs are drawn from distributions, not measured from time and resource use.
- Each payer pays a fixed multiple of Medicare by service, with no patient cost sharing detail.
- Referrals into other services, quality, outcomes, philanthropy and oral cancer drugs are excluded.

With real data, replace the generator with remittance and claim files, electronic health record encounters and the cost accounting extract, map the columns to the same schema, and rerun the integrity check first.

## Resume bullets and skills

Combined version. Keep "synthetic data" in the project line so no one reads the figures as hospital results.

1 Built an end-to-end Python analysis of cancer service line profitability on synthetic data (3,600 patients, about 80,000 encounters), producing an encounter-level P&L from gross charge to operating margin ($210.7M net revenue, $17.0M margin).

2 Showed that the margin depends on commercial payers (+$27.7M against −$10.6M for all other payers combined) and on one 340B drug discount worth $17.7M; without it the line loses $0.6M.

3 Wrote an integrity test that recomputes the P&L from raw rows and ties every output table to it, so every chart and figure reconciles.

4 Compared four overhead allocation methods and showed cancer type rankings change with the method (prostate −2.1% to 6.4%), which argues against cutting a service line on one number.

5 Trained a gradient boosting denial-risk model with a patient-grouped split; the top-20% review list caught 43% of denials against 19% for random review (2.2 times the precision) and was worth $1.7M a year, or $3.0M when ranked by expected dollars.

6 Reported that the model matched a simple payer-and-service baseline (AUC 0.704 against 0.706) and recommended starting with the rule until claim-level features are available.

7 Stress tested the P&L under 11 scenarios, including loss of 340B (−$17.7M), site-neutral payment (−$5.2M), a 5% commercial rate increase (+$5.3M) and denial recovery (+$2.7M).

8 Delivered 11 charts, an interactive dashboard, a 13-slide deck and a written case study, with a seeded, reproducible pipeline and a README with setup instructions.

### Skills highlighted

| | |
|---|---|
| **Programming** | Python, pandas, NumPy, scikit-learn, matplotlib; seeded, reproducible pipelines; automated integrity checks |
| **Healthcare finance** | Service line P&L, payer contracting, revenue cycle and denials, 340B drug economics, cost allocation, site-neutral payment, scenario analysis |
| **Machine learning** | Gradient boosting classification, patient-grouped train and test split, baseline comparison, precision at top-k, worklist value in dollars |
| **Communication** | Data visualization, dashboards, executive slide deck, case study writing, stating limits of synthetic data |

## Run it

Requires Python 3.10 or later. From the project folder (the one containing `README.md`):

```
python -m venv .venv
source .venv/bin/activate        # Windows: .venv\Scripts\Activate.ps1
pip install -r requirements.txt

python src/generate_data.py --patients 3600 --seed 42 --outdir data
python src/analyze.py --data data --out outputs
python tests/check_integrity.py   # ends with "All integrity checks passed."
python src/denial_model.py --data data --out outputs
python src/make_charts.py         # writes 11 PNGs to charts/
```

The same seed reproduces the same data, results and charts. The full instructions and troubleshooting are in the README.

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All figures are from synthetic data, fiscal 2025, and demonstrate a method. They are not hospital benchmarks.